Recall

Work from memory, then check with your tutor.

R1. Write a story for $\frac{1}{6} \div 3$ and solve it.

Answer: _____

R2. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

Answer: _____

R3. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

Answer: _____

R4. A double number line pairs 3 minutes with 8 laps. What pairs with 12 minutes?

Answer: _____

S. Return here near the end of the lesson

Before using a percent, I decide which amount represents ___.

Percent as a rate per hundred

Follow the slide labels. P = problem; F = your turn; D = discussion.

P1. Find 10%, 25%, 50%, and 75% of 80 counters.

Answer: _____

P2. A shelf holds 40 books. Twelve are mysteries. What percent are mysteries?

Answer: _____

P3. Twenty-four stickers are 40% of a collection. How many stickers are in the whole collection?

Answer: _____

Percent as a rate per hundred

Follow the slide labels. P = problem; F = your turn; D = discussion.

P4. Find 125% of 48 minutes.

Answer: _____

P5. Find 35% of 60 tiles.

Answer: _____

P6. Find 15% of $18.00.

Answer: _____

Percent as a rate per hundred

Follow the slide labels. P = problem; F = your turn; D = discussion.

F. Twelve coins are 20% of a jar. Ten percent is __ coins. One hundred percent is __ coins.

Answer: _____

D. Omar says 24 is 40% of 9.6 because 0.40 × 24 = 9.6. Explain the error.

Answer: _____

Exit ticket

Work on your own. Show a model or reason when the question asks for one.

E. Twenty-one marbles are 35% of a bag. How many marbles are in the bag?

Answer: _____

Name _____ Date _____ Week 3

Mini Check A

15 minutes. No calculator. Show your work.

1. Compare $\frac{2}{9}$ and $\frac{7}{9}$.

Answer: _____

2. Which is closer to 1: $\frac{7}{8}$ or $\frac{4}{5}$?

Answer: _____

3. A picture shows 5 green tiles and 2 white tiles. Write green : white and white : green.

Answer: _____

4. Is 6 : 10 equivalent to 9 : 15?

Answer: _____

5. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

Answer: _____

Mini Check A

6. Pack A has 4 erasers for $1.80. Pack B has 6 for $2.40. Which costs less per eraser?

Answer: _____

7. Find 140% of 30 points.

Answer: _____

8. Find 35% of 80 tiles.

Answer: _____

Rates and percents

The big idea

A unit rate describes an amount per one unit. Percent means per hundred. Label the whole before deciding whether to find a part, a whole, or a percent.

Words to know

Rate: A ratio comparing quantities with different units. Example: 15 kilometers in 3 hours.

Unit rate: An amount per one unit of another quantity. Example: 5 kilometers per hour.

Per: For each one. Example: $0.60 per apple means $0.60 for one apple.

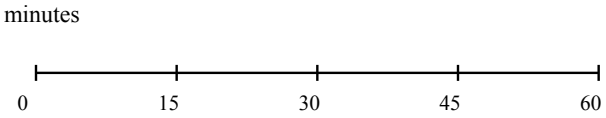
Percent: A rate per hundred. Example: 35% means 35 for every 100.

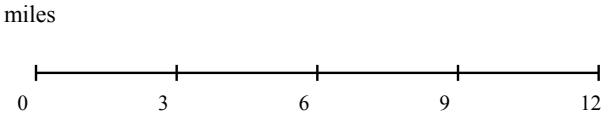
Whole: The amount that represents 100%. Example: If all 80 tiles count, 80 is the whole.

Benchmark percent: A familiar percent useful for reasoning. Example: 25% is one fourth.

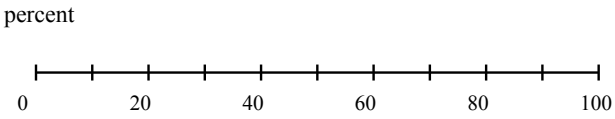
How it works

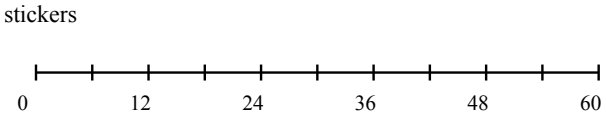
Four 15-minute intervals make one hour. A unit rate names how much of one quantity goes with one unit of another. Its units show the division order.





100% labels the unknown whole. Percent compares an amount with a whole split into 100 equal parts.





Worked examples

1. A walker travels 2 miles in 30 minutes. Find miles per hour.

60 minutes is two groups of 30.

Two groups of 2 miles make 4 miles.

The rate is 4 miles per hour.

Answer: 4 miles per hour

2. Eighteen blocks are 30% of a tower set. Find the whole set.

30% has three 10% groups.

18 ÷ 3 = 6 blocks per 10%.

Ten groups give 10 × 6 = 60 blocks.

Answer: 60 blocks

Common mistakes

Divides in the reverse direction. Why: Does not track the per unit. Fix: Read the desired rate aloud before dividing.

Treats 20 minutes as 0.20 hours. Why: Uses base ten instead of 60 minutes per hour. Fix: Partition an hour into three 20-minute sections.

Multiplies the known part by the percent. Why: Treats every percent problem as finding a part. Fix: Place the known part and its percent on aligned lines.

Try it

1. A bus covers 10 miles in 25 minutes at constant speed. How many miles in one hour?

2. Pack A has 4 erasers for $1.80. Pack B has 6 for $2.40. Which costs less per eraser?

3. Find 140% of 30 points.

4. Find 35% of 80 tiles.

5. Find 12% of $15.00.

Answers

1. 24 miles 2. Pack B: $0.40 each 3. 42 points 4. 28 tiles 5. 1.8000 dollars

Where this connects

This builds on equivalent ratios. Fraction division later extends the same grouping and scaling ideas.